SLEEP 08

Ibutamoren (MK-677)

Small non-peptide molecule taken by mouth that mimics ghrelin



CLASS Small molecule (GH secretagogue)

STUDIED FOR GH, aging, hip fracture

HUMAN EVIDENCE Controlled human trials

US STATUS Not FDA approved · FDA Category 2

STUDIED FOR

- Older adults — HUMAN
- Alzheimer's disease — HUMAN
- Hip fracture recovery — HUMAN
- Child GH deficiency — HUMAN
- Sleep — HUMAN

One hip fracture trial was stopped. It was stopped early for a safety signal of congestive heart failure; FDA cites this in its Category 2 listing.

HOW IT WORKS IN THE BODY

Ibutamoren ▶ GHS receptor ▶ Growth hormone ▶ IGF-1 rises

- **Mimics ghrelin** at the growth hormone secretagogue receptor
- **Pituitary releases** more growth hormone, raising IGF-1
- **IGF-1 rose** 73% to 84% in the Alzheimer's and hip fracture trials
- **For sleep,** the effect rests on one small 1997 study

EVIDENCE & WHAT'S KNOWN

EVIDENCE

CONTROLLED TRIALS
LEVEL 4 OF 5

- Placebo RCTs in older adults and patients
- Sleep: one study, 14 people (1997)
- Phase III in children recruiting (2026)

NOT YET ESTABLISHED

- Effect on sleep in larger trials
- Long-term safety
- Half-life
- Drug interactions

QUICK PRACTICAL INFO

HALF-LIFE Not established

ROUTES STUDIED By mouth (trials)

PHARMACOKINETICS Limited human data

SOLD AS "Research use only" (not legal for human use)

MONITORED IN TRIALS GH, IGF-1, glucose, insulin sensitivity, function

SPORT (WADA) Prohibited, class S2

REGULATORY

FDA	Not approved
FDA	Category 2 (503A, 503B): heart failure risk
WADA	Prohibited (S2.2.4)

REFERENCES

1. FDA, Category 2 bulk drug substances (significant safety risks), Apr 2026
2. Nass et al., Ann Intern Med, ibutamoren in older adults, Nov 2008
3. Copinschi et al., Neuroendocrinology, MK-677 and sleep, Oct 1997

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